LECTURE NOTES

Topic 2: Problem Analysis

Learning Outcomes:

(a) Define the Input-Process-Output (IPO) model **(1 hour)**

LECTURE NOTES

RECAP :

- Problem analysis is the first step in problem solving process.
- Problem analysis is the act of identifying **input**, **process** and **output (IPO)**.
- Programmers use **IPO model to** organize and summarize the results of the given problem.

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IPO Model

The Input-Process-Output (IPO) Model is an approach to describe and visualize the input, process and output to solve the given problem.



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Problem Analysis

Problem analysis is the activities of identifying

- **input**
- **process and/or condition (if any)**
- **output**

from the problem scenario.

Example of problem scenarios:

- **Find the area of a rectangle given width and height**
- **Find the area of a circle**
- **Calculate the sum of two numbers**

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IPO Model Description

INPUT	PROCESS	OUTPUT
Input is the value entered/inserted by user	Process is the manipulation (arithmetic/ logical comparison) of input to produce output. Print is not a process.	Output is the expected result asked in the problem statement/ question.
Keywords: input, enter, read, get, from the problem statements	**Keywords**: calculate, compute, count, determine, find, check, compare, if, else, or, otherwise, repeat, times, while, loop, until, as long as from the problem statements	**Keywords** : display, print, show, convey, find from the problem statements
represented by a descriptive name (noun)	Represented by a verb that involves sequence, selection and repetition control structure process	represented by a descriptive name (noun) or message (“ “)
Eg: name, number, price, width, height, weight	**Eg: calculate the sum of two numbers, calculate the area of a rectangle based on length and width, determine the average mark based on two quizzes, repeat calculating total mark of 10 students**	**Eg: sum, total, area, average, “Pass”, “Hello world”**

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Steps to create IPO Model

1. Analyze the given problem
2. Determine the goal – Output
3. Determine the items needed to achieve that goal – Input
4. Determine activity that takes place to achieve the goal – Process

- Always search **first** for the **Output**.

LECTURE NOTES

IPO Model

Calculate and display sum of three numbers.		
INPUT	**PROCESS**	**OUTPUT**
number1, number2, number3	Calculate Sum based on number1, number2 and number3 or Calculate Sum based on three numbers	Sum

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Conclusion

Create an IPO model (*refer example below*)

Find the area of a rectangle given width and height		
INPUT	**PROCESS**	**OUTPUT**
width, height	**Calculate area of rectangle based on width and height**	**area of rectangle**

Problem statement

IPO chart / diagram

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Topic 2: Problem Analysis

Learning Outcomes:

2.3 Apply problem analysis to determine the input, process, and output (IPO) for given scenarios using selection control structures. **(2 hours)**

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Topic 2: Problem Analysis

Selection Control Structure

Decide which statement(s) to execute based on a condition.

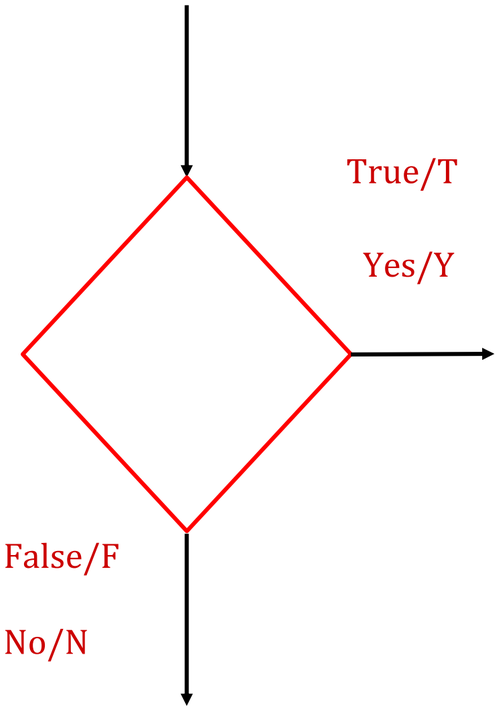
Defines two courses of action depending on the outcome of a condition.

(example : Print "Pass" or "Fail" based on grade entered)

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Topic 2: Problem Analysis

- The selection control structure performs an action(s) based on a certain condition
- In pseudocode, selection will be represented using the **if** statement
- In the flowchart, the selection will be represented using decision symbol (represented by diamond shape)



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Topic 2: Problem Analysis

There are three (3) types of selection control structure:

1. Single Selection
2. Dual Selection
3. Multiple Selection

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Topic 2: Problem Analysis

In **Single Selection**, the if statement performs an action if the specified condition (Boolean expression) is true. If the condition is false, nothing is done.

if statement :

Checks a condition to perform action(s)
If condition is True (T), perform actions(s)
If condition is False (F), skips it (i.e. perform nothing)

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Topic 2: Input, Process, Output

IPO Analysis General Format	
Input	List all required *input,* separated by **comma**
Process	**Determine** *output* based on *input* **Explanation of how to get output based on the **input** entered* **No formula in process*
Output	*TrueOutput **or** no output*

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Topic 2: Problem Analysis

Example 1 : The program will display “A” if mark entered is greater or equal to 80

IPO Analysis
Input : *mark* Process : Determine ***“A”*** based on mark OR Determine message based on mark Output :***“A”*** *or no output*

Note :
For single selection, please mention “message / value *or no output*”

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Topic 2: Problem Analysis

In **Dual Selection**, if the Boolean expression evaluates to true, the statement(s) for the true case is executed; otherwise, the statement(s) for the false case are executed instead.

if … else statement :

Check one (1) condition to choose between two (2) actions
If the condition is True (T), perform action1, otherwise (if condition is False (F)), perform action2

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Topic 2: Problem Analysis

IPO Analysis General Format	
Input	List all required *input,* separated by **comma**
Process	**Determine** *TrueOutput or FalseOutput* based on *input* **No formula in process*
Output	*TrueOutput* ***or*** *FalseOutput*

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Topic 2: Problem Analysis

Example 2 : The program will display grade “A” if the mark entered is greater or equal to 80 or grade “B” if the mark entered is less than 80

IPO Analysis
Input : *mark* Process : Determine ***“A”*** *or* ***“B”*** based on *mark* or Determine ***grade*** based on *mark*   Output :***“A”*** *or* ***“B”***

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Topic 2: Problem Analysis

In **Multiple Selection**. conditions will be checked one by one, when a condition evaluates to true, perform the true case and stop checking the rest.

if … else if…else statement :

Checks many conditions to choose between many actions
Condition will be checked one by one
When a condition is True, perform the action and stop checking the rest.

Note : exclude nested selection

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Topic 2: Problem Analysis

IPO Analysis General Format	
Input	List all required *input,* separated by **comma**
Process	**Determine** (write all possible *output)* based on *input* **No formula in process*
Output	List all possible output, separated by the word or

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Topic 2: Problem Analysis

Example 3: The program will display grade “A” if mark entered is greater or equal to 80 or grade “B” if mark entered is in between 70 to 79 or grade “C” if mark entered is less than 70

IPO Analysis
Input : *mark* Process : Determine ***“A”*** *or* ***“B” or “C”*** based on *mark* or Determine ***grade*** based on *mark* Output :***“A”*** *or* ***“B”*** *or* ***“C”***

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Topic 2: Problem Analysis

Exercise 1: Identify the input, process and output for the given problem statement.

Print the message “You are entitled to vote” for a person whose age is above 18

Input :

Process :

Output :

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Topic 2: Problem Analysis

Exercise 1: Identify the input, process and output for the given problem statement.

Print the message “You are entitled to vote” for a person whose age is above 18

Input : age

Process : Determine “You are entitled to vote” based on age

Output : “You are entitled to vote” *or no output*

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Topic 2: Problem Analysis

Exercise 2: Identify the input, process and output for the given problem statement.

A program will calculate age of a person's. If the age of the person is above 55, then the program will print “Retired”, otherwise, the program will print “Still working”.

Input :

Process :

Output :

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Topic 2: Problem Analysis

Exercise 2: Identify the input, process and output for the given problem statement.

A program will calculate age of a person's. If the age of the person is above 55, then the program will print “Retired”, otherwise, the program will print “Still working”.

Input : age

Process : Determine “Retired” or “Still working” based on age

Output : “Retired” or “Still working”

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Topic 2: Problem Analysis

Exercise 3: Identify the input, process and output for the given problem statement.

The program will compare two numbers and display either message “num1 is greater than num2” or message “num2 is greater than num1”

Input :

Process :

Output :

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Topic 2: Problem Analysis

Exercise 3: Identify the input, process and output for the given problem statement.

The program will compare two numbers and display either message “num1 is greater than num2” or message “num2 is greater than num1”

Input : number1, number2

Process : Determine “num1 is greater than num2” or “num2 is greater than num1” based on number1, number2

Output : “num1 is greater than num2” or “num2 is greater than num1”

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Topic 2: Problem Analysis

Exercise 4: Identify the input, process and output for the given problem statement.

User enters weight and height. Calculate BMI and print “Overweight” if BMI i> 25.0 display **“Normal”** if the BMI is between **18.0 and 25.0 (inclusive)**, otherwise display **“Underweight”**

Input :

Process :

Output :

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Topic 2: Problem Analysis

Exercise 4: Identify the input, process and output for the given problem statement.

User enters weight and height. Calculate BMI and print “Overweight” if BMI i> 25.0 display **“Normal”** if the BMI is between **18.0 and 25.0 (inclusive)**, otherwise display **“Underweight”**

Input : weight, height

Process : Calculate BMI based on weight, height.
Determine “Overweight” or “Normal” or “Underweight” based on BMI

Output : “Overweight” or “Normal” or “Underweight”

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Topic 2: Problem Analysis

Learning Outcomes:

(d) Apply problem analysis to determine the input, process, and output (IPO) for given scenarios using repetition control structures.**(2 hours)**

Repetition also known as looping, iteration

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Topic 2: Problem Analysis

Specifies a block of one or more statements that are repeatedly executed until a condition is satisfied

2 types of repetition :

1. Counter-controlled
2. Sentinel-controlled (A value that indicates the end of repetition process)

   E.g- “Do you want to continue(Y/N)?”

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Topic 2: Problem Analysis (Approach 1)

In **Counter-Controlled Repetition**, the number of loops that need to be executed is known (either specified in the question or determined by the user), and it depends on the value specified in the condition.

IPO Analysis General Format	
Input	**input for _____ times**
Process	**Repeat calculate *output* based on input for _____ times**
Output	***output* for _____ times**

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Topic 2: Problem Analysis (Approach 2)

In **Counter-Controlled Repetition**, the number of loops that need to be executed is known (either specified in the question or determined by the user), and it depends on the value specified in the condition.

IPO Analysis General Format	
Input	**input**
Process	**Repeat calculate *output* based on input for ____ times**
Output	***output***

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Topic 2: Problem Analysis (Approach 1)

Example 1 : The program calculate BMI for 5 persons

IPO Analysis
Input : *weight*, *height* **for 5 times** Process : **Repeat** read *weight* and *height,* **calculate BMI based on** *weight* and *height* **for 5 times** Output : **BMI for 5 times**

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Topic 2: Problem Analysis (Approach 2)

Example 1 : The program calculate BMI for 5 persons

IPO Analysis
Input : *weights*, *heights* Process : **Repeat calculate BMI based on** *weights* and *heights* **for 5 times** Output : **BMI**

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Topic 2: Problem Analysis

Sentinel-controlled repetition is sometimes called indefinite repetition because the number of loops to be executed is not known in advance.

It is a repetition procedure for solving a problem by using a sentinel value (also called a signal value, a dummy value or a flag value) to indicate "end of data entry".

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Topic 2: Problem Analysis (Approach 1)

IPO Analysis General Format	
Input	input for x times until _____
Process	Repeat calculate *output* based on input until _____
Output	*output* for x times until _____

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Topic 2: Problem Analysis (Approach 2)

IPO Analysis General Format	
Input	**input**
Process	**Repeat calculate *output* based on input until _____**
Output	***output***

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Topic 2: Problem Analysis (Approach 1) -tukar soalan lain

Example 2 : The program calculates BMI of persons and stops when the user enters 0.

IPO Analysis
Input : ***weight, height*** **for *x times* until user enter 0** Process : **Repeat** calculate **BMI based on *weight, height* until user enter 0** Output : **BMI for *x times* until user enter 0**

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Topic 2: Problem Analysis (Approach 2) tukar soalan lain

Example 2 : The program calculates BMI of persons and stops when the user enters 0.

IPO Analysis
Input : ***weight, height*** Process : **Repeat** calculate **BMI** **based on weight and height until user enter 0** Output : **BMI**

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Topic 2: Problem Analysis

Exercise 1 : The program will calculate sum and average of 5 numbers

Input :

Process :

Output :

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Topic 2: Problem Analysis

Exercise 2: Program will calculate and print the average for a few numbers entered by user and terminate when 0 is entered

Input :

Process :

Output :

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Topic 2: Problem Analysis

Exercise 3: This program allows users to repeatedly enter numbers for 10 times. The program will only display numbers that are divisible by 5 and calculate its sum values.

Input :

Process :

Output :

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Topic 2: Problem Analysis

Exercise 4: Write a program that will accept the age of 10 persons. If the age of the person is above 55, then the program will print “Retired”, otherwise, the program will print “Still working”.

Input :

Process :

Output :

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Topic 2: Problem Analysis

Identify the type of control structure based on the following statements.

Statement	Control Structure
i) Sam wants a program that displays the number of seconds it takes for a baseball to travel a specified distance at a specified speed.	
ii) Mr. Tan has a car rental business in Langkawi. Customers will pay a base fee according to the size of car rented. Small size (S) is RM50, medium size (M) is RM80 and large size (L) is RM100 per day.	
iii) The staff of HEP Unit need to find the total of module 1, module 2 and module 3 students who have attended the Local Rule Seminar. The program should also find the percentage of attendent for each module. The total of the participants are 100 students.	